

Subject:

Date:

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Full Subtractor design by NAND & NOR gate

• Truth Table of Full Subtractor

A	B	C	Difference	Borrow
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$$\text{Difference} = A'B'C + A'BC' + AB'C' + ABC$$

$$\text{Borrow} = A'B'C + A'BC' + A'BC + ABC$$

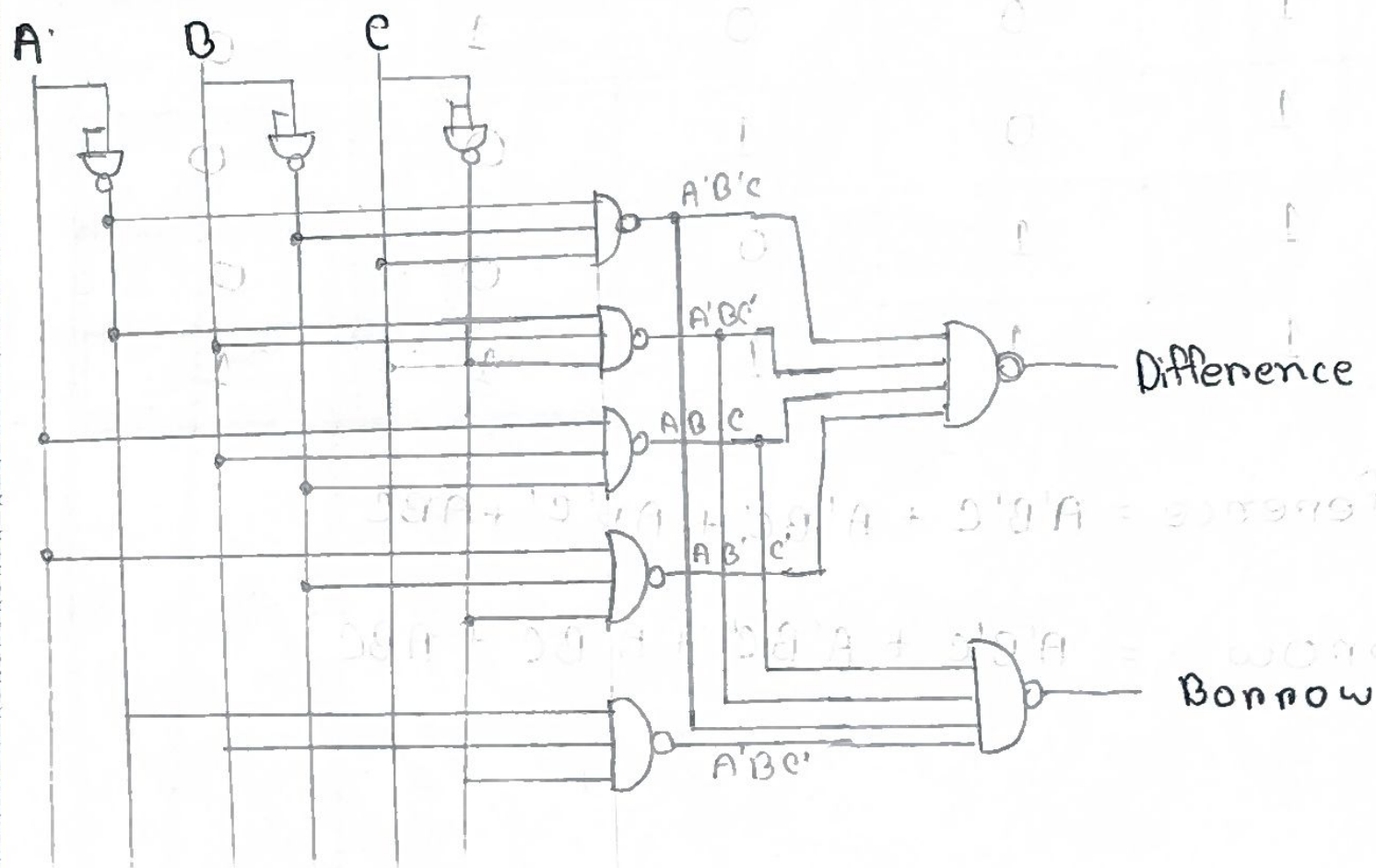
NAND Gate

$$\text{Difference} = \overline{A'B'C} + \overline{A'BC'} + \overline{AB'C'} + \overline{ABC}$$

$$= (\overline{A'B'C}) (\overline{A'BC'}) (\overline{AB'C'}) (\overline{ABC})$$

$$\text{Borrow} = \overline{A'B'C} + \overline{A'BC'} + \overline{A'BC} + \overline{ABC}$$

$$= (\overline{A'B'C}) (\overline{A'BC'}) (\overline{A'BC}) (\overline{ABC})$$



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NOR Gate

$$\text{Difference} = A'B'C + A'BC' + AB'C' + ABC$$

$$= (A+B+C') (A+B'+C) (A'+B+C) (A'+B'+C')$$

$$= (\overline{A+B+C'}) + (\overline{A+B'+C}) + (\overline{A'+B+C}) + (\overline{A'+B'+C'})$$

$$\text{Borrow} = A'B'C + A'BC' + A'BC + ABC$$

$$= (A+B+C') (A+B'+C) (A+B'+C') \cdot (A'+B'+C')$$

$$= (\overline{A+B+C}) + (\overline{A+B'+C}) + (\overline{A+B'+C'}) + (\overline{A'+B'+C'})$$

